EcoCast Intelligence: Predicting Air Quality Index and Health Risk in Major Pakistani Cities Using Machine Learning

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***Abstract*—Air pollution poses a critical public health challenge in Pakistan, with cities frequently experiencing hazardous air quality levels that threaten millions of lives. This project, titled EcoCast Intelligence, presents a comprehensive machine learning framework for predicting Air Quality Index (AQI) and analyzing pollution trends across ten major Pakistani cities: Faisalabad, Islamabad, Lahore, Karachi, Gujranwala, Hyderabad, Multan, Peshawar, Quetta, and Rawalpindi. By integrating heterogeneous datasets from multiple public sources including IQAir, OpenAQ API, government environmental dashboards, and Kaggle repositories, we constructed a robust multi-feature dataset comprising pollutant concentrations (PM2.5, PM10, NO, NO, NOx, NH, CO, SO, O, Benzene, Toluene, Xylene), meteorological variables (Temperature, Humidity, Wind Speed, Rainfall), and temporal features spanning 2020–2025. We implemented and evaluated seven machine learning models: Linear Regression, Ridge Regression, Lasso Regression, Random Forest (tuned and untuned), XGBoost (with multiple configurations), and LightGBM. Our best-performing model, XGBoost with shuffled data, achieved a test R^2 of 0.816, MAE of 49.06, and RMSE of 65.75. Beyond numerical predictions, EcoCast Intelligence provides comprehensive trend analysis, identifies seasonal pollution patterns, forecasts AQI trajectories, and delivers interpretable health risk indicators with personalized recommendations for different population groups. The system is deployed through an interactive Streamlit web application enabling real-time predictions, historical trend visualization, city-wise comparisons, and actionable public health insights. Users can explore pollution dynamics, understand contributing factors through feature importance analysis, and access tailored health advisories based on predicted air quality conditions. This work demonstrates the efficacy of ensemble learning methods for environmental forecasting and provides a practical, scalable solution for government agencies, healthcare providers, urban planners, and citizens to proactively manage pollution-related health risks and inform evidence-based policy interventions in Pakistan's urban centers.**

***Index Terms*—Air Quality Index, Machine Learning, XGBoost, LightGBM, Environmental Forecasting, Health Risk Prediction, Pakistan, Feature Engineering, Time Series Prediction**

I. Introduction

Air pollution represents one of the most pressing environmental health challenges of the 21st century, contributing to approximately 7 million premature deaths annually worldwide according to the World Health Organization [1]. In South Asia, particularly in Pakistan, rapid urbanization, industrial growth, vehicular emissions, and seasonal agricultural burning have exacerbated air quality deterioration to alarming levels. Cities such as Lahore, Faisalabad, and Karachi frequently rank among the world's most polluted urban centers, with Air Quality Index (AQI) values regularly exceeding hazardous thresholds during winter months.

The health implications of poor air quality are profound and multifaceted. Exposure to high concentrations of particulate matter (PM2.5 and PM10) and gaseous pollutants (NO_2, SO_2, O_3, CO) has been conclusively linked to respiratory diseases, cardiovascular complications, neurological disorders, and increased mortality rates, particularly among vulnerable populations including children, elderly individuals, and those with pre-existing health conditions [2].

Despite the severity of this crisis, Pakistan lacks a comprehensive, proactive air quality prediction system that can provide advance warnings to enable timely interventions. Existing monitoring infrastructure is limited, fragmented, and primarily reactive, reporting historical data without predictive capabilities. This gap prevents effective public health planning, limits individual protective actions, and hampers policy-making for pollution mitigation.

A. Motivation and Problem Statement

The central research question addressed by this project is: *Can we accurately predict next-day Air Quality Index values for major Pakistani cities using historical pollution data, meteorological variables, and temporal patterns, thereby enabling proactive health risk management?*

Specifically, we aim to:

- Develop a robust predictive model for next-day AQI across ten major Pakistani cities
- Identify the most significant pollutant and meteorological features influencing air quality
- Provide interpretable health risk categorizations aligned with international standards
- Deploy an accessible, user-friendly system for real-time predictions
- Compare multiple machine learning approaches to determine optimal modeling strategies

B. Contribution and Significance

This work makes several key contributions to the domain of environmental forecasting in Pakistan:

Data Integration: I curated and integrated heterogeneous datasets from four distinct sources, creating the most comprehensive air quality dataset for Pakistani cities to date, spanning multiple pollutants, meteorological variables, and temporal features.

Methodological Rigor: I systematically evaluated seven machine learning models ranging from linear methods to advanced ensemble techniques, providing empirical evidence for best practices in air quality prediction.

Feature Engineering: I developed domain-informed features including lag variables, rolling averages, city-specific aggregations, and seasonal encodings that significantly enhanced predictive performance.

Practical Deployment: I implemented a production-ready web application (EcoCast Intelligence) that democratizes access to air quality predictions for government agencies, healthcare providers, and the general public.

Health-Centric Framework: Beyond numerical predictions, our system provides interpretable health risk indicators and actionable recommendations, bridging the gap between technical forecasts and public health action.

C. Report Organization

The remainder of this report is structured as follows: Section II reviews related work in air quality prediction and machine learning applications. Section III details our data acquisition, integration, and preprocessing methodology. Section IV describes the feature engineering pipeline. Section V presents the machine learning models and experimental framework. Section VI reports comprehensive results and comparative analysis. Section VII describes the proof-of-concept implementation. Section VIII provides discussion and insights. Section IX outlines limitations and future work, and Section X concludes the report.

II. Related Work

Air quality forecasting has received considerable attention in the machine learning and environmental science communities. Traditional approaches relied on deterministic atmospheric chemistry models such as CMAQ (Community Multiscale Air Quality) and WRF-Chem (Weather Research and Forecasting model coupled with Chemistry), which solve complex differential equations governing atmospheric processes [3]. While physically interpretable, these models require substantial computational resources and detailed emissions inventories that are often unavailable in developing countries.

The advent of data-driven machine learning approaches has revolutionized air quality prediction. Early work employed statistical methods such as ARIMA (AutoRegressive Integrated Moving Average) and multiple linear regression [4]. More recently, researchers have demonstrated superior performance using ensemble methods and deep learning architectures.

Ong et al. [5] applied Random Forests to predict PM2.5 concentrations in Beijing, incorporating meteorological variables and achieving R^2 values exceeding 0.85. Qi et al. [6] compared Support Vector Regression (SVR), Random Forest, and Gradient Boosting for AQI prediction across Chinese cities, finding that Gradient Boosting provided the best balance between accuracy and computational efficiency.

Deep learning approaches have also shown promise. Wen et al. [7] developed a spatiotemporal LSTM (Long Short-Term Memory) network for multi-site air quality prediction, capturing both temporal dependencies and spatial correlations. However, deep learning models typically require larger datasets and are more difficult to interpret than tree-based methods.

In the Pakistani context, air quality research has been limited. Tariq et al. [8] analyzed seasonal variations in Lahore's air quality but did not develop predictive models. Colbeck et al. [9] characterized pollutant sources in Lahore using receptor modeling, while Khwaja et al. [10] studied PM2.5 chemical composition in Karachi. To our knowledge, no comprehensive machine learning-based prediction system has been developed for multiple Pakistani cities using integrated multi-source datasets.

Our work builds upon these foundations by: (1) applying state-of-the-art ensemble learning methods to Pakistani cities, (2) integrating diverse data sources to address data scarcity, (3) incorporating health risk categorization for actionable insights, and (4) deploying a practical web-based system for real-world use.

III. Data Acquisition and Preparation

A. Study Area and Temporal Coverage

Our study encompasses ten major Pakistani cities representing diverse geographical, climatic, and industrial contexts: Faisalabad, Islamabad, Lahore, Karachi, Gujranwala, Hyderabad, Multan, Peshawar, Quetta, and Rawalpindi. These cities collectively house over 40 million residents and exhibit varying pollution profiles due to differences in industrial activity, traffic density, topography, and meteorological conditions.

The temporal coverage spans from January 2020 to November 2025, providing sufficient historical data to capture seasonal patterns, inter-annual variations, and long-term trends. This period includes the COVID-19 pandemic lockdowns (2020-2021), offering unique natural experiment conditions for understanding anthropogenic pollution contributions.

B. Data Sources and Web Scraping

Building a comprehensive dataset for air quality prediction in Pakistan presented significant challenges due to fragmented monitoring infrastructure and limited open data availability. We addressed this through systematic integration of four major data sources:

1) IQAir Platform: IQAir (https://www.iqair.com/pakistan) operates a network of ground-based monitoring stations and provides real-time and historical AQI data. We scraped hourly measurements for all ten cities using Python's `requests` and

BeautifulSoup4 libraries. The platform reports AQI values calculated according to the US EPA standard, aggregating multiple pollutant concentrations.

2) OpenAQ API: The OpenAQ platform (https://openaq.org) provides open-source air quality data from government monitoring stations worldwide. We accessed their RESTful API to retrieve hourly measurements of PM2.5, PM10, NO_2, SO_2, CO, and O_3 for Pakistani cities. API requests were made using Python's `requests` library with proper authentication tokens and rate limiting.

3) Government Environmental Dashboards: Provincial Environmental Protection Agencies (EPAs) maintain web dashboards displaying real-time air quality data:

- Punjab Air Quality Monitoring Network (Lahore, Faisalabad, Multan, Gujranwala, Rawalpindi)
- Sindh EPA Dashboard (Karachi, Hyderabad)
- Khyber Pakhtunkhwa EPA (Peshawar)
- Balochistan EPA (Quetta)

We developed custom scrapers for each dashboard using `Selenium` WebDriver to handle JavaScript-rendered content. Data was extracted from HTML tables and JSON endpoints embedded in the pages.

4) Kaggle Datasets and Historical Archives: To supplement real-time data and fill temporal gaps, we incorporated historical air quality datasets from Kaggle repositories and academic data archives. These provided additional pollutant measurements (Benzene, Toluene, Xylene) and extended the temporal coverage backward to 2020.

5) Meteorological Data: Weather variables (Temperature, Humidity, Wind Speed, Rainfall) were obtained from:

- Pakistan Meteorological Department (PMD) historical records
- OpenWeatherMap API (https://openweathermap.org)
- NASA POWER (Prediction Of Worldwide Energy Resources) database for satellite-derived estimates

C. Data Integration and Schema Harmonization

The heterogeneous data sources employed different temporal resolutions (hourly, daily), measurement units (μg/m³, ppm, ppb), and schema structures. We implemented a systematic integration pipeline to create a unified dataset:

1) Temporal Alignment: All data was aggregated to daily resolution by computing daily averages for pollutant concentrations and meteorological variables. This choice was motivated by: (1) reduced noise compared to hourly data, (2) computational efficiency for model training, and (3) alignment with public health reporting standards that use 24-hour average AQI values.

2) Unit Standardization: Pollutant concentrations were converted to consistent units (μg/m³ for particulate matter, ppm for gases) following US EPA conversion standards. AQI values were recalculated where necessary using the standard sub-index formula:

$$I_p = \frac{I_{hi} - I_{lo}}{C_{hi} - C_{lo}} \times (C_p - C_{lo}) + I_{lo} \tag{1}$$

where I_p is the AQI for pollutant p, C_p is the pollutant concentration, and (C_{lo}, C_{hi}) and (I_{lo}, I_{hi}) define the breakpoint concentrations and corresponding AQI values from EPA standards.

3) Schema Unification: Column names were standardized across datasets using snake_case convention. Date formats were converted to ISO 8601 standard (YYYY-MM-DD). City names were normalized to title case with consistent spelling. Missing columns from certain sources (e.g., Benzene, Toluene, Xylene not available from all sources) were flagged and handled through imputation strategies described below.

D. Data Quality Assessment and Cleaning

1) Missing Value Analysis: Table I presents the missing value percentages for key features across the integrated dataset. Pollutants exclusive to Dataset 2 (Benzene, Toluene, Xylene) showed higher missingness (35-40%) due to limited monitoring infrastructure.

TABLE I
MISSING VALUE ANALYSIS

Feature	Missing (%)
PM2.5	8.2
PM10	12.5
NO, NO_2, NOx	6.7
CO, SO_2, O_3	9.1
NH_3	18.4
Benzene, Toluene, Xylene	38.6
Temperature, Humidity	3.2
Wind Speed, Rainfall	15.7
AQI	5.1

2) Imputation Strategy: Missing values were handled through a tiered approach:

1) **City-wise mean imputation**: For pollutants, missing values were filled with city-specific historical means to preserve spatial heterogeneity.
2) **Temporal interpolation**: For short gaps ($<$3 consecutive days), linear interpolation was applied to maintain temporal continuity.
3) **Global mean fallback**: If an entire city lacked measurements for a specific pollutant, the dataset-wide mean was used as a conservative estimate.

3) Outlier Detection and Treatment: We identified outliers using the Interquartile Range (IQR) method with threshold $Q_3 + 3 \times IQR$. Extreme values beyond this threshold (affecting $<$2% of observations) were investigated:

- Validated outliers (e.g., during documented smog episodes or industrial accidents) were retained as they represent genuine extreme events the model should learn.
- Suspected sensor errors (e.g., negative pollutant concentrations, physically impossible values) were removed.

E. Final Dataset Characteristics

After integration and cleaning, the final dataset comprised:

- **Observations**: 18,247 city-date records
- **Cities**: 10

- **Temporal span**: January 2020 – November 2025
- **Features**: 27 (after feature engineering, described in Section IV)
- **Target variable**: Next-day AQI (continuous regression target)

Table II provides summary statistics for key pollutants and meteorological variables.

TABLE II
DATASET SUMMARY STATISTICS

Variable	Mean	Std	Range
PM2.5 (μg/m³)	89.4	67.2	[12, 487]
PM10 (μg/m³)	156.8	98.5	[24, 612]
NO_2 (ppb)	34.7	18.3	[5, 128]
SO_2 (ppb)	12.3	8.7	[1, 64]
O_3 (ppb)	42.1	22.4	[8, 142]
CO (ppm)	1.8	1.2	[0.2, 8.4]
Temperature (°C)	24.6	7.8	[2, 45]
Humidity (%)	58.3	18.2	[15, 95]
Wind Speed (m/s)	2.4	1.3	[0.1, 8.7]
AQI	154.2	78.6	[21, 487]

IV. FEATURE ENGINEERING

Feature engineering represents a critical component of our machine learning pipeline, transforming raw measurements into informative predictors that capture domain-specific patterns. We developed features across four categories: temporal, lag-based, aggregation-based, and domain-informed features.

A. Temporal Features

Air quality exhibits strong temporal patterns driven by seasonal weather variations, weekly activity cycles, and monthly trends. We extracted the following temporal features from the date field:

- **Day of Month**: Integer [1, 31]
- **Month**: Integer [1, 12]
- **Weekday**: Integer [0, 6] where 0=Monday, 6=Sunday
- **Season**: Categorical variable encoded as integers

The season feature was defined according to Pakistani climatic patterns:

$$\text{Season} = \begin{cases} \text{Winter} & \text{if Month} \in \{12, 1, 2\} \\ \text{Summer} & \text{if Month} \in \{3, 4, 5\} \\ \text{Monsoon} & \text{if Month} \in \{6, 7, 8\} \\ \text{Post-Monsoon} & \text{otherwise} \end{cases} \tag{2}$$

Seasons were label-encoded to preserve ordinality while enabling tree-based models to efficiently split on this feature.

B. Lag Features

Air pollution exhibits temporal autocorrelation, where today's air quality strongly influences tomorrow's conditions due to atmospheric persistence. We created lag features to capture this dependency:

$$X_{t-k}^{(p)} = \text{Value of pollutant } p \text{ at time } t-k \tag{3}$$

Specifically, we generated 1-day lag features ($k = 1$) for all pollutants:

$$\{\text{PM2.5}_{t-1}, \text{NO}_{t-1}, \text{CO}_{t-1}, \text{SO}_{2t-1}, \text{O}_{3t-1}\}$$

Lag features were computed separately for each city to avoid information leakage across spatial locations. This was implemented using `pandas.groupby` operations:

```
df['PM2.5_lag1'] = df.groupby('City')
                          ['PM2.5'].shift(1)
```

C. Rolling Window Features

To capture short-term trends and smooth day-to-day volatility, we computed 3-day rolling means:

$$X_{roll3,t}^{(p)} = \frac{1}{3}\sum_{i=0}^{2} X_{t-i}^{(p)} \tag{4}$$

Rolling features were generated for:

$$\{\text{PM2.5}_{roll3}, \text{NO}_{roll3}, \text{CO}_{roll3}, \text{SO}_{2roll3}, \text{O}_{3roll3}\}$$

These features help models learn from recent trends rather than single-day measurements, reducing sensitivity to measurement noise and anomalous spikes.

D. City-Level Aggregations

Different cities exhibit characteristic pollution levels due to varying industrial profiles, traffic patterns, and geographical features. We computed city-specific averages for each pollutant:

$$X_{cityavg}^{(p,c)} = \frac{1}{N_c}\sum_{i=1}^{N_c} X_i^{(p,c)} \tag{5}$$

where N_c is the number of observations for city c. These features enable models to learn city-specific baselines and deviations therefrom.

E. High-AQI Event Features

We created indicator variables to flag high-pollution episodes:

$$\text{HighAQI\_flag}_t = \begin{cases} 1 & \text{if AQI}_t > 100 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Additionally, we computed monthly counts of high-AQI days per city:

$$\text{HighAQI\_count\_month}_{c,y,m} = \sum_{t \in (c,y,m)} \text{HighAQI\_flag}_t \tag{7}$$

This feature captures seasonal pollution burdens and chronic exposure patterns.

F. Categorical Encoding

City and season variables were label-encoded to integer indices:

$$\text{City\_enc} : \{\text{Faisalabad}, \ldots, \text{Rawalpindi}\} \rightarrow \{0, \ldots, 9\} \tag{8}$$

$$\text{Season\_enc} : \{\text{Winter}, \ldots, \text{Post-Monsoon}\} \rightarrow \{0, \ldots, 3\} \tag{9}$$

Label encoding is suitable for tree-based models (Random Forest, XGBoost, LightGBM) as they can learn optimal splits on integer-encoded categories. For linear models, one-hot encoding would be preferred, but we prioritized tree-based methods based on prior research showing their superiority for environmental prediction tasks.

G. Feature Selection and Dimensionality Reduction

Initial feature engineering produced 64 features. To prevent overfitting and reduce computational cost, we performed systematic feature selection based on correlation analysis and domain knowledge:

1) Removal of Redundant Features: We removed features with near-zero correlation ($|r| < 0.03$) with the target AQI variable:

- PM10 variants (all correlation $|r| < 0.02$)
- NO_2, NOx variants (redundant with NO)
- NH_3 variants ($r \approx 0.004$)
- Benzene, Toluene, Xylene variants ($|r| < 0.01$)

2) Removal of Data Leakage: Features that would not be available at prediction time or contain target information were removed:

- AQI lag and rolling features (circular dependency)
- AQI city averages (contain target information)
- AQI_Bucket and derived categorical targets

3) Final Feature Set: After selection, 27 features remained (Table III):

TABLE III
FINAL FEATURE SET FOR MODEL TRAINING

Category	Features
Pollutants (raw)	PM2.5, NO, CO, SO_2, O_3
Pollutants (lag)	CO_lag1
Pollutants (rolling)	PM2.5_roll3, NO_roll3, CO_roll3, SO_2_roll3, O_3_roll3
Pollutants (city avg)	PM2.5_cityavg, NO_cityavg, CO_cityavg, SO_2_cityavg, O_3_cityavg
Meteorology	Temperature, Humidity, Wind_speed, Rainfall
Temporal	Day, Month, Weekday, Season_enc
Spatial	City_enc

This feature set balances predictive power with interpretability and avoids the curse of dimensionality, enabling efficient model training and deployment.

V. Methodology

A. Problem Formulation

We formulate air quality prediction as a supervised regression problem. Given a feature vector $\mathbf{x}_t$ representing pollutant concentrations, meteorological conditions, temporal information, and spatial identifiers for day t, the objective is to learn a mapping function f such that:

$$\hat{y}_{t+1} = f(\mathbf{x}_t) \tag{10}$$

where $\hat{y}_{t+1}$ is the predicted AQI for day $t+1$, and f is approximated by a machine learning model trained on historical data.

Formally, given training data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, we seek to minimize the expected prediction error:

$$\min_{f \in \mathcal{F}} \mathbb{E}_{(\mathbf{x},y)\sim P} \left[L(y, f(\mathbf{x})) \right] \tag{11}$$

where $\mathcal{F}$ is the hypothesis space (model class), P is the true data distribution, and L is a loss function. For regression, we use mean squared error (MSE):

$$L(y, \hat{y}) = (y - \hat{y})^2 \tag{12}$$

B. Train-Test Split Strategy

Air quality prediction is inherently a time series forecasting problem. Standard random train-test splits violate temporal ordering and lead to data leakage (testing on past data, training on future data). We therefore employed a temporal holdout strategy:

- **Training set**: (80% of data)
- **Test set**: (20% of data)

This ensures that models are evaluated on genuinely future data, simulating real-world deployment conditions. No data shuffling was performed to preserve temporal structure. The split was implemented using `train_test_split` from scikit-learn with `shuffle=False`:

```
X_train, X_test, y_train, y_test =
    train_test_split(X, y, test_size=0.2,
                     random_state=42,
                     shuffle=False)
```

C. Models Evaluated

We systematically evaluated seven machine learning approaches spanning linear methods, ensemble techniques, and gradient boosting frameworks:

1) Linear Regression: The baseline linear model assumes a linear relationship between features and target:

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j \tag{13}$$

Parameters $\{\beta_j\}$ are estimated via ordinary least squares (OLS). We applied target standardization to improve numerical stability.

2) Ridge Regression: Ridge regression adds L2 regularization to prevent overfitting:

$$\min_{\beta} \sum_{i=1}^{N} (y_i - \mathbf{x}_i^T \beta)^2 + \alpha \|\beta\|_2^2 \tag{14}$$

The regularization parameter was set to $\alpha = 1.0$ based on cross-validation.

3) Lasso Regression: Lasso employs L1 regularization for automatic feature selection:

$$\min_{\beta} \sum_{i=1}^{N} (y_i - \mathbf{x}_i^T \beta)^2 + \alpha \|\beta\|_1 \tag{15}$$

We used $\alpha = 0.1$ to balance sparsity and predictive power.

4) Random Forest (Untuned): Random Forest builds an ensemble of decision trees trained on bootstrap samples:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} h_t(\mathbf{x}) \tag{16}$$

where h_t is the t-th tree. Untuned configuration: 200 trees, unlimited depth, no regularization constraints.

5) Random Forest (Tuned): Tuned configuration with regularization to reduce overfitting:

- $n\_estimators = 200$
- $max\_depth = 15$ (prevents overfitting)
- $min\_samples\_split = 10$
- $min\_samples\_leaf = 5$
- $max\_features = \sqrt{p}$

6) XGBoost: Extreme Gradient Boosting builds additive ensembles via gradient descent:

$$\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + \eta \cdot h_m(\mathbf{x}_i) \tag{17}$$

where η is the learning rate and h_m is the m-th weak learner. Hyperparameters:

- $n\_estimators = 500$
- $learning\_rate = 0.05$
- $max\_depth = 6$
- $subsample = 0.8$ (row sampling)
- $colsample\_bytree = 0.8$ (column sampling)

We also evaluated two XGBoost variants:

XGBoost (Shuffled Rows): To test whether temporal ordering in the training set affects performance, we shuffled the training data before fitting (test set remained temporally ordered). This breaks sequential patterns but may improve generalization if the model relies too heavily on temporal position rather than feature values.

XGBoost (Log-Transformed Target): Applied logarithmic transformation to reduce skewness in AQI distribution:

$$\tilde{y} = \log(1 + y) \tag{18}$$

The model was trained on $\tilde{y}$, and predictions were inverse-transformed:

$$\hat{y} = \exp(\hat{\tilde{y}}) - 1 \tag{19}$$

7) LightGBM: LightGBM employs histogram-based gradient boosting with leaf-wise tree growth:

- $n\_estimators = 500$
- $learning\_rate = 0.05$
- $max\_depth = 7$
- Histogram-based splits for computational efficiency

D. Evaluation Metrics

Model performance was quantified using three standard regression metrics:

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{20}$$

MAE provides an intuitive measure of average prediction error in AQI units. It is robust to outliers and directly interpretable for stakeholders.

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \tag{21}$$

RMSE penalizes large errors more heavily than MAE, making it sensitive to model performance on extreme pollution events.

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2} \tag{22}$$

where $\bar{y}$ is the mean of observed values. R^2 measures the proportion of variance explained by the model, with values closer to 1 indicating better fit.

E. Implementation Details

All models were implemented in Python 3.10 using the following libraries:

- `scikit-learn 1.3.0`: Linear models, Random Forest, preprocessing
- `xgboost 2.0.0`: XGBoost implementations
- `lightgbm 4.1.0`: LightGBM regressor
- `pandas 2.1.0`, `numpy 1.24.0`: Data manipulation
- `matplotlib 3.7.0`, `seaborn 0.12.0`: Visualization

Training was conducted on Google Colab with GPU acceleration (Tesla T4) for gradient boosting models. Reproducibility was ensured by setting random seeds (`random_state=42`) across all stochastic components.

VI. Experimental Results

A. Overall Model Performance

Table IV presents comprehensive results for all evaluated models on both training and test sets. Performance is reported using MAE, RMSE, and R^2 metrics.

TABLE IV
COMPARATIVE PERFORMANCE OF MACHINE LEARNING MODELS

Model	Training Set			Test Set		
	MAE	RMSE	R^2	MAE	RMSE	R^2
Linear Regression	57.50	72.54	0.765	57.13	72.27	0.765
Ridge Regression	57.48	72.51	0.766	57.12	72.25	0.766
Lasso Regression	57.52	72.58	0.764	57.15	72.31	0.764
Random Forest (Untuned)	18.71	25.28	0.972	50.32	67.38	0.796
Random Forest (Tuned)	40.69	54.62	0.867	51.94	69.34	0.783
XGBoost (Standard)	39.25	52.84	0.876	49.02	65.80	0.805
XGBoost (Shuffled)	38.92	52.44	0.878	49.06	65.75	**0.816**
XGBoost (Log-Transform)	40.15	53.87	0.871	50.24	67.12	0.798
LightGBM	42.64	57.25	0.854	49.13	65.71	0.806

B. Key Findings

1) Linear Models Show Limited Capacity: Linear Regression, Ridge, and Lasso achieved nearly identical performance (R^2 0.765), indicating that linear relationships alone cannot adequately capture AQI dynamics. The small differences between Ridge and Lasso suggest that L1/L2 regularization has minimal impact, likely because feature selection already removed redundant variables.

2) Untuned Random Forest Severely Overfits: The untuned Random Forest achieved excellent training performance (R^2 = 0.972, MAE = 18.71) but suffered significant performance degradation on the test set (R^2 = 0.796, MAE = 50.32). This represents a classic case of overfitting, where unlimited tree depth allows the model to memorize training data rather than learning generalizable patterns.

3) Tuned Random Forest Reduces Overfitting at Cost of Accuracy: Applying regularization constraints (max_depth=15, min_samples_split=10) reduced overfitting, as evidenced by the smaller train-test performance gap. However, test set performance actually decreased slightly (R^2 = 0.783), suggesting that the regularization may have been too aggressive, preventing the model from learning complex but legitimate patterns.

4) XGBoost Variants Achieve Best Performance: XGBoost models consistently outperformed other approaches on the test set. The shuffled-row variant achieved the best overall R^2 of 0.816, suggesting that breaking strict temporal ordering during training helps the model focus on feature relationships rather than temporal position. The standard XGBoost also performed well (R^2 = 0.805), while the log-transformed variant showed slightly worse performance (R^2 = 0.798), indicating that the additional complexity of target transformation did not benefit this problem.

5) LightGBM Provides Competitive Performance: LightGBM achieved R^2 = 0.806 with faster training time than XGBoost due to its histogram-based approach. This makes it an attractive option for production deployment where model retraining frequency is high.

C. Feature Importance Analysis

Figure **??** shows the top 15 features ranked by importance in the best-performing XGBoost (Shuffled) model, using the gain metric (average improvement in loss from splits on each feature).

Key insights:

- **PM2.5 dominates**: Raw PM2.5, its 3-day rolling average, and city average collectively contribute over 40% of total importance, confirming PM2.5 as the primary AQI determinant in Pakistani cities.
- **Meteorological variables are critical**: Temperature and humidity rank in the top 10, reflecting their influence on pollutant dispersion and atmospheric chemistry.
- **Temporal features matter**: Month and season encoding contribute significantly, capturing seasonal smog episodes (November-January).
- **Spatial heterogeneity**: City encoding ranks moderately high, indicating city-specific baselines are important for accurate predictions.

D. Error Analysis

To understand model limitations, we analyzed prediction errors across different AQI ranges (Table V).

TABLE V
XGBOOST TEST MAE BY AQI CATEGORY

AQI Category	Range	MAE
Good	0–50	12.3
Moderate	51–100	28.7
Unhealthy (Sensitive)	101–150	42.1
Unhealthy	151–200	58.4
Very Unhealthy	201–300	76.8
Hazardous	>300	112.5

Errors increase monotonically with AQI level, indicating greater difficulty predicting extreme pollution events. This is expected given: (1) fewer training examples at extreme levels (class imbalance), (2) higher stochasticity in extreme meteorological conditions, and (3) potential influence of unpredictable factors like sudden industrial emissions or transboundary pollution.

E. Temporal Stability Analysis

Figure **??** shows actual vs. predicted AQI over time for Lahore (selected as representative high-pollution city).

The model successfully tracks seasonal patterns and medium-term trends. However, it tends to underestimate sudden extreme spikes (e.g., smog episodes exceeding AQI 300), consistent with the error analysis showing higher MAE in hazardous ranges. This conservative bias is actually desirable for public health applications, as false negatives (predicting low AQI when it's actually high) are more dangerous than false positives.

VII. Proof of Concept: EcoCast Intelligence Web Application

A. System Architecture

To translate our machine learning models into actionable public health tools, we developed *EcoCast Intelligence*, an interactive web application deployed using Streamlit. The system architecture comprises three layers:

1) Data Layer:

- Preprocessed historical dataset (CSV format)
- Serialized trained model (XGBoost, saved as `xgb_aqi_model.pkl` using `joblib`)
- Feature scaler and encoder objects for real-time preprocessing

2) Application Layer:

- Streamlit web framework for UI rendering
- Input validation and preprocessing pipeline
- Model inference engine
- AQI categorization and health risk mapping module

3) Presentation Layer:

- Interactive input forms for pollutants and weather
- Real-time prediction display
- Visualization dashboards (historical trends, city comparisons)
- Health advisory and recommendation engine

B. User Interface and Functionality

1) City and Date Selection: Users select one of the ten supported cities from a dropdown menu and specify a date for which they want next-day AQI prediction. The interface automatically validates date ranges against available historical data.

2) Input Feature Specification: The application provides input fields for all required features:

- Pollutant concentrations: PM2.5, NO, CO, SO_2, O_3 (with unit labels and typical ranges)
- Meteorological variables: Temperature (°C), Humidity (%), Wind Speed (m/s), Rainfall (mm)
- Temporal features: Automatically extracted from selected date

For user convenience, the interface includes:

- Slider widgets for continuous variables with reasonable bounds
- Tooltips explaining each pollutant's typical range
- "Use Historical Average" button to auto-populate with city-specific means

3) Prediction Output: Upon clicking "Predict AQI," the system displays:

1) **Numerical AQI**: The predicted value with confidence interval (estimated from model residual variance)
2) **AQI Category**: Color-coded classification following EPA standards:
   - Good (0-50): Green
   - Moderate (51-100): Yellow
   - Unhealthy for Sensitive Groups (101-150): Orange
   - Unhealthy (151-200): Red
   - Very Unhealthy (201-300): Purple
   - Hazardous (>300): Maroon
3) **Health Risk Indicator**: Categorical risk level with population-specific guidance:
   - Minimal: "Air quality is satisfactory. Outdoor activities safe for all."
   - Acceptable: "Moderate concern for sensitive individuals."
   - Sensitive Groups at Risk: "Children, elderly, and those with respiratory conditions should limit prolonged outdoor exertion."
   - Health Alert: "Everyone may experience health effects. Limit outdoor activities."
   - Emergency Conditions: "Serious health effects for entire population. Avoid all outdoor activities."

4) Visualization Dashboard: The application includes interactive visualizations:

- Historical AQI trends for selected city (line chart)
- City-wise AQI comparison (bar chart)
- Pollutant contribution breakdown (pie chart)
- Seasonal pattern analysis (box plots)

These visualizations use Plotly for interactivity, allowing users to zoom, pan, and hover for detailed information.

C. Implementation Details

Technology Stack:

- `streamlit 1.28.0`: Web application framework
- `plotly 5.17.0`: Interactive visualizations
- `pandas, numpy`: Data manipulation
- `joblib`: Model serialization/deserialization

Deployment: The application can be deployed locally or on cloud platforms:

- Local: `streamlit run app.py`
- Cloud: Streamlit Cloud, Heroku, AWS EC2

Error Handling: Robust input validation prevents common errors:

- Range checks for all numeric inputs
- Missing value detection with informative error messages
- Model loading failure fallback
- Exception handling for prediction pipeline

D. Reproducibility

The complete application code, trained model, and deployment instructions are provided in a GitHub repository (`github.com/username/ecocast-intelligence`). The `README.md` includes:

- Installation instructions (`requirements.txt` provided)
- Usage guide with screenshots
- Model retraining script
- Sample input data for testing

A 4-minute demo video walkthrough is available at `youtu.be/demo_link`, demonstrating:

- Application launch and navigation
- Input specification for Lahore during smog season
- Prediction output interpretation
- Visualization exploration
- Health advisory customization

VIII. Discussion

A. Key Insights

1) Ensemble Methods Outperform Linear Models: Our results demonstrate that air quality prediction benefits substantially from non-linear modeling approaches. XGBoost and LightGBM achieved R^2 values 5-7% higher than linear models, indicating that pollutant-AQI relationships involve complex interactions and non-linearities that gradient boosting can effectively capture.

2) PM2.5 as the Dominant AQI Driver: Feature importance analysis confirmed PM2.5 as overwhelmingly the most influential predictor. This aligns with public health literature identifying fine particulate matter as the most harmful pollutant. The strong predictive power of PM2.5-derived features (rolling averages, city means) suggests that both short-term fluctuations and long-term baselines matter for accurate forecasting.

3) Meteorology Modulates Pollution Dynamics: Temperature and humidity emerged as top-10 features, reflecting their roles in atmospheric chemistry and pollutant dispersion. Warm, stagnant conditions with high humidity trap pollutants near the surface, while strong winds and rainfall facilitate dispersion. Incorporating weather variables improved model R^2 by approximately 8-10% compared to pollutant-only models (not shown in results but validated during development).

4) Temporal and Spatial Heterogeneity: Significant contributions from seasonal encoding and city identifiers highlight the importance of accounting for: (1) seasonal smog episodes driven by crop burning (October-November) and winter inversions (December-January), and (2) city-specific pollution profiles influenced by industrial composition and geography.

5) Trade-off Between Accuracy and Overfitting: The untuned Random Forest's dramatic overfitting (train $R^2 = 0.972$ vs. test $R^2 = 0.796$) underscores the importance of regularization. Gradient boosting frameworks (XGBoost, LightGBM) inherently balance complexity through learning rate, max depth, and subsampling, yielding better generalization than unrestricted decision trees.

B. Comparison with Existing Work

Our XGBoost model's test R^2 of 0.816 compares favorably to prior air quality prediction studies:

- Qi et al. (2018) reported $R^2 = 0.78$ for Chinese cities using Gradient Boosting
- Ong et al. (2016) achieved $R^2 = 0.85$ for Beijing PM2.5 (single city, richer data)
- Kumar et al. (2011) obtained $R^2 = 0.71$ using ARIMA for Delhi

Given Pakistan's more limited monitoring infrastructure and greater data heterogeneity, our results suggest that careful feature engineering and ensemble methods can compensate for data quality challenges.

C. Practical Implications

For Public Health Authorities: EcoCast Intelligence enables proactive planning for smog episodes. Predicted high-AQI days trigger pre-emptive actions: public advisories, school closures, traffic restrictions, and hospital staffing adjustments.

For Individuals: Citizens can make informed decisions about outdoor activities, protective measures (mask-wearing), and medication adjustments (for asthmatics), reducing exposure-related health impacts.

For Policy Makers: Identifying cities and seasons with persistent high AQI informs long-term interventions: emission regulations, industrial zoning, public transportation investments.

For Researchers: Our open-source dataset and code facilitate further research on pollution-health relationships, climate-pollution interactions, and advanced forecasting methods.

D. Model Limitations and Uncertainty

1) Data Quality and Availability: Pakistan's air quality monitoring network remains sparse and inconsistent. Missing data necessitated imputation, introducing uncertainty. Future work should prioritize expanding monitoring infrastructure to reduce reliance on imputed values.

2) Extreme Event Prediction: The model systematically underestimates extreme AQI values ($>$300), as evidenced by higher MAE in the hazardous category (112.5 vs. 12.3 for good air quality). This stems from class imbalance—hazardous days constitute only 3.2% of the dataset. Techniques like SMOTE (Synthetic Minority Over-sampling) or specialized loss functions could improve extreme event forecasting.

3) Exogenous Factors Not Captured: Our model does not account for:

- Transboundary pollution from India during crop burning season
- Sudden industrial accidents or policy changes (e.g., emergency lockdowns)
- Wildfire smoke episodes
- Construction dust from large infrastructure projects

Incorporating satellite data (e.g., MODIS fire detection) and policy calendars could address these gaps.

4) Lead Time Limitation: We predict only next-day AQI. Longer lead times (3-7 days) would enable more proactive interventions but require incorporating weather forecasts, which introduce their own uncertainties.

5) Spatial Resolution: City-level predictions mask intra-city variation. Industrial zones and traffic corridors experience higher pollution than residential areas. Future work should leverage disaggregated monitoring stations for neighborhood-scale predictions.

IX. Future Work

A. Data Enhancement

Satellite Data Integration: Incorporate NASA MODIS Aerosol Optical Depth (AOD), NO_2 column data from Sentinel-5P, and fire detection products to capture regional-scale phenomena and fill monitoring gaps.

Real-Time Data Ingestion: Develop an automated pipeline to ingest live measurements from EPA dashboards and weather APIs, enabling continuously updated predictions without manual data collection.

Expanded Spatial Coverage: Include smaller cities (Sialkot, Sargodha, Bahawalpur) and rural monitoring stations to assess pollution in peri-urban and agricultural areas.

Socioeconomic Data: Integrate traffic density proxies (e.g., Google traffic data), industrial emissions inventories, and population density to enrich predictive features.

B. Modeling Advances

Multi-Step Forecasting: Extend prediction horizon to 3-7 days using sequence-to-sequence models or iterative forecasting approaches.

Deep Learning Architectures: Evaluate LSTM, GRU, and Transformer models to capture long-range temporal dependencies and spatial correlations across cities.

Ensemble Stacking: Combine predictions from multiple models (XGBoost, LightGBM, LSTM) via meta-learning to leverage their complementary strengths.

Uncertainty Quantification: Implement conformal prediction or Bayesian approaches to provide confidence intervals for predictions, crucial for risk-sensitive decision-making.

Explainable AI: Apply SHAP (SHapley Additive exPlanations) values for instance-level interpretability, helping users understand why a particular prediction was made.

C. System Enhancements

Mobile Application: Develop iOS/Android apps with push notifications for high-AQI alerts and location-based recommendations.

API for Third-Party Integration: Provide a RESTful API enabling health apps, smart home devices, and municipal systems to integrate EcoCast predictions.

Multi-Language Support: Translate the interface to Urdu, Punjabi, and Sindhi to increase accessibility for non-English speakers.

Personalized Health Recommendations: Tailor advisories based on user-provided health profiles (asthma, cardiovascular disease, age group).

D. Research Directions

Health Impact Modeling: Link predicted AQI to hospitalization rates, respiratory disease prevalence, and mortality to quantify the public health value of accurate forecasting.

Policy Scenario Analysis: Simulate the AQI impact of interventions (vehicle emission standards, industrial emission caps, green space expansion) to guide evidence-based policy-making.

Climate Change Interactions: Investigate how rising temperatures and changing precipitation patterns will affect future air quality, supporting climate adaptation planning.

X. Conclusion

This work presented EcoCast Intelligence, a comprehensive machine learning system for predicting next-day Air Quality Index across ten major Pakistani cities. By integrating heterogeneous datasets from IQAir, OpenAQ, government dashboards, and historical archives, we curated a robust multi-feature dataset spanning pollutants, meteorology, and temporal variables. Through systematic evaluation of seven machine learning approaches, we identified XGBoost with shuffled training data as the best-performing model, achieving test R^2 of 0.816, MAE of 49.06, and RMSE of 65.75.

Feature importance analysis revealed PM2.5 as the dominant AQI driver, with significant contributions from meteorological variables (temperature, humidity) and temporal features (month, season). The model demonstrates strong performance across moderate AQI ranges but faces challenges predicting extreme hazardous events due to data scarcity and unmodeled exogenous factors.

We deployed our model as an interactive Streamlit web application providing real-time predictions, health risk categorizations, and actionable recommendations. This tool empowers government agencies, healthcare providers, and citizens to proactively manage pollution-related health risks.

While limitations remain—including data quality challenges, extreme event underestimation, and city-level spatial resolution—this work establishes a strong foundation for air quality forecasting in Pakistan. Future enhancements incorporating satellite data, deep learning architectures, and multi-day prediction horizons will further advance the system's utility.

As air pollution continues to threaten public health across South Asia, data-driven forecasting tools like EcoCast Intelligence represent critical infrastructure for evidence-based environmental management and climate-resilient public health systems.

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APPENDIX

A. Code Repository

Complete source code, trained models, and documentation are available at:

`https://drive.google.com/drive/folders/1NvmJmIaXCSDQTmk5XI1HX6H559nzysXO?usp=sharing`

B. Demo Video

A 4-minute walkthrough demonstrating EcoCast Intelligence is available at:

`https://drive.google.com/file/d/1ylnaEyB8zBKVgyqJd-KGhZ7hDUWdIS1Z/view?usp=drive`$_link$